

Livsförsäkringsmatematik II Projekt 1

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1 Lost in data.

For project one, we are given 5 different assignments to solve based on a data set. So, let's first dive into the data!

The data comprises 4398 individuals and their development through 10 years. The values reported are 0 which represents death, 1 is sick and 2 being healthy, we interchangeably throughout this project use the term state to describe 0, 1 and 2. As the reader can suspect, if an individual dies, they cannot transition to sick or healthy (unfortunately, once you die, you stay dead). An individual can transition freely between sick and healthy and both sick and healthy individuals can die.

We start this adventure by calculating how many days in total are spent in each state, see table 1.

Tabell 1: Total days spent in state.

0 = Dead	1 = Sick	2 = Healthy
50704	502396	15499600

From table 1 we can clearly see that the majority of time spent by the total population is in the healthy state, the obvious conclusion from this is that death given the data is rare. An interesting question to ask is, is our data accurate/ representative? The data used in this project is based upon 4398 50-year-old individuals and their journey towards age 60. A way to detect anything peculiar would be to compare the data against Statistiska centralbyrån in short SCB. SCB is responsible for official statistics and other government statistics in Sweden. According to SCB (2023-12-22) year 2022 there was a total of 135095 50 years old. With the use of the total population of 50 years old in Sweden, we get that the proportion of the data compared to the total population is $\frac{4398}{135095} \approx 3.3\%$. How many of the 4398 individuals die under 10 years? We can easily calculate this by counting the number

of zeros found in the last column, equivalent to the last day of the data set. We get that out of 4398 individuals, 35 have died under 10 years. This is a percentage of $\frac{35}{4398} \approx 0.8\%$. The first death occurred on day 135, and the last on day 3641 (unlucky). We would expect the number of deaths to increase as time goes by. To examine this, we aggregate the data from days to years, counting the number of deaths given the specific year, the results shown in table 2.

Year	Number of deaths
1	2
2	1
3	6
4	4
5	0
6	1
7	6
8	5
9	3
10	8

Tabell 2: Deaths per year.

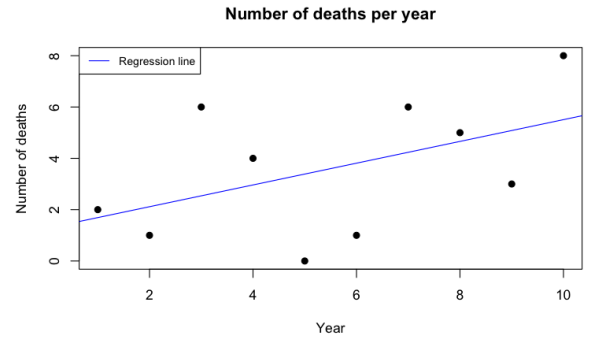


Figure 1

From Table 2, it looks like the number of deaths increases as the population gets older. But a really interesting find is that at year 5 no deaths are reported. Figure 1 boosts our confidence of a year-by-year increase in dying. Let's repeat this procedure with data from SCB.

Year	Deaths per year (SCB).
1	259
2	251
3	256
4	346
5	361
6	430
7	474
8	481
9	528
10	565

Tabell 3: Deaths per year.

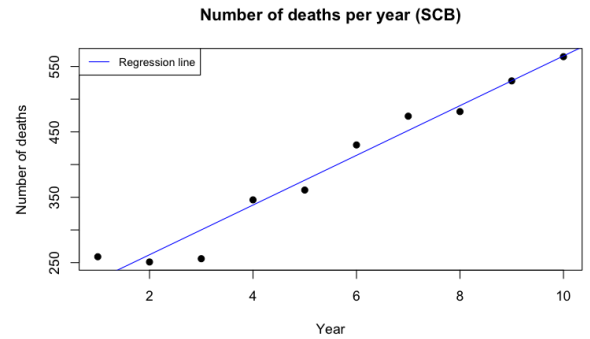


Figure 2

From Table 3, we can see an increase in deaths every 2 to 3 years. From Figure 2 it surely looks like a steep increase in the number of deaths year by year.

The previous tables and figures can be misleading depending on the population size at year

50. Now, we know that the sizes of each population are different. The number of individuals in our data is 4398 and SCB 135095. So, to come to a conclusion about if the data we are given is realistic or approximately realistic we will divide the number of deaths per year with the respective number of 50 year old alive at age 50, multiply this by 100 and we get the year by year percentage deaths, see table 4.

Year	Our data	SCB
1	0.05	0.19
2	0.02	0.19
3	0.14	0.19
4	0.09	0.26
5	0	0.27
6	0.02	0.32
7	0.14	0.35
8	0.11	0.36
9	0.07	0.39
10	0.18	0.42

Tabell 4: Percentage Deaths per year given total pop. at year 50.

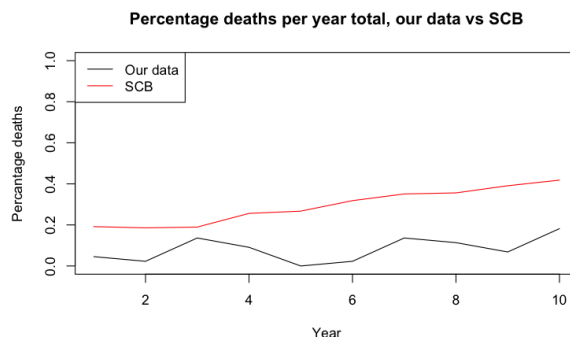


Figure 3

Clearly, the data we are given underestimates the year-by-year percentage number of deaths. Figure 4 shows us that according to SCB, the number of deaths between ages 50 and 60 in the population is low, 0.2% to 0.8%, and from Figure 3, it looks like it's increasing linearly. The reader can compare our figure with SCB (2023-12-22, ref: 2).

1.1 Summary

We could not find equivalent data on sickness. Data about sickness from SCB is on individuals who also work and can be reported as sick because of sick leave. Another major issue is that the data we found aggregates age. We wonder if this is done to reduce discrimination based on age. We would assume that the distribution of the rate of becoming sick is bell-shaped, telling us that younger and older individuals get sick more often, but this could be false. If the hypothesis of a bell-shaped distribution were true, it would be very interesting to find where ages 50 to 60 lie. These limitations are the reason for only studying death.

The data exploration has shown us that our data set does not seem to match the actual level of deaths in the population of 50 to 60-year-olds. From this discovery, we need to raise a major concern. If the data we use to estimate different actuarial estimates under reports deaths, it is likely that our estimates will be affected. So, how could we possibly tackle this major issue? By bootstrapping our data. This will be covered in Assignment 5.

2 Assignment 1

In modeling interest rates, we are given a table of yearly spot-rates. What we want to accomplish is the calculation of the interest rate intensity $r(t)$ given the spot-rates r_t . We can do this using the following steps. The yearly discount factor is given by (1).

$$D(0, t) = e^{-(t \cdot r_t)} \quad (1)$$

We can rewrite (1) as,

$$D(0, t) = \exp\left(-\int_0^t r(s) ds\right) = \exp(-t \cdot r_t). \quad (2)$$

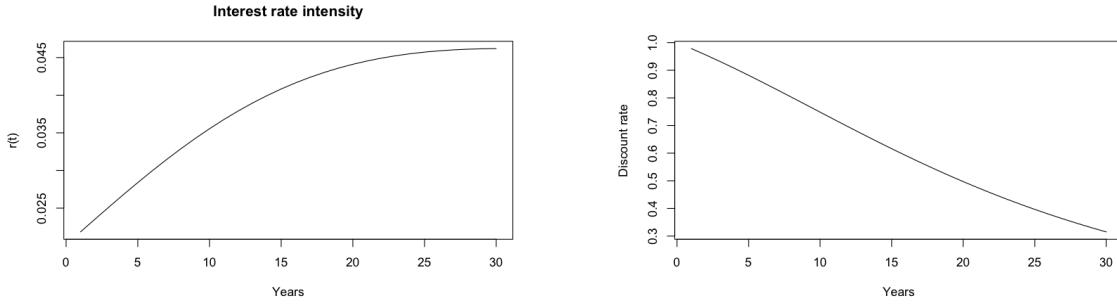
Solving for $r(t) \approx \tilde{r}_t$ we can simplify (2) as,

$$\exp\left(-\sum_{s=1}^t \tilde{r}_s\right) = \exp(-t \cdot r_t) \longleftrightarrow \sum_{s=1}^t \tilde{r}_s = t \cdot r_t. \quad (3)$$

From (3) we can calculate the interest rate intensity as follows,

$$\begin{aligned} \sum_{s=1}^t \tilde{r}_s &= t \cdot r_t \\ \tilde{r}_{s=1} &= 1 \cdot r_1 \\ \tilde{r}_{s=2} &= 2 \cdot r_2 - \tilde{r}_{s=1} \\ \tilde{r}_{s=3} &= 3 \cdot r_3 - \tilde{r}_{s=1} - \tilde{r}_{s=2} \\ &\vdots \\ \tilde{r}_{s=n} &= n \cdot r_n - \sum_{j=1}^{n-1} \tilde{r}_j \end{aligned} \quad (4)$$

Which is what we will use throughout this project. Plotting $D(0, t)$ as a function of t and interest rate intensity $r(t)$, you get the results shown in Figure 4 below.



(a) Interest rate intensity for the spot rates.

(b) Discount rate as a function of the spotrates.

3 Assignment 2

It is now time for us to estimate the transition rates for transitions between the states 2 and 1, as functions of age (this only contains very few observed deaths). We can do this by following the step-by-step guide in Norberg R. (2002) chapter 11. Statistical inference in the Markov chain model. The estimated transition rates are calculated with (5), based upon equation 11.61 Norberg.R (2002).

$$\hat{\mu}_{gh,i} = \frac{N_{gh,i}}{W_{g,i}} \quad (5)$$

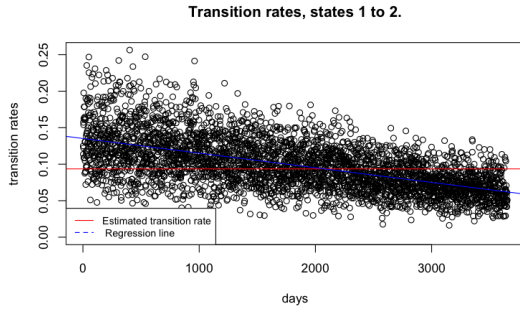
We have that $i = \text{day}$, $N_{gh,i}$ = The number of jumps from $g \rightarrow h$ under day i and $W_{g,i}$ = total time spent in state g under day i . We create a mammoth of a code that calculates (5). The first function calculates $N_{gh,i}$ and $W_{g,i}$ and after we have counted $N_{gh,i}$ and $W_{g,i}$ the second function calculate the intensities.

We calculate the transition matrix, which gives us the estimated $\hat{\mu}_{g,h}(t)$ for the entire duration t .

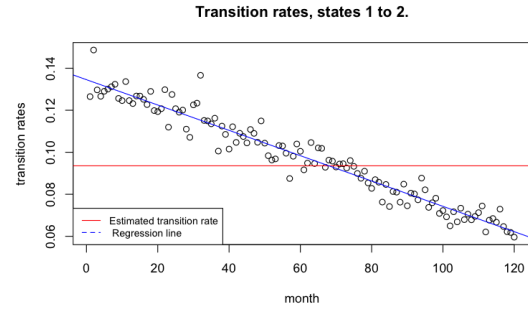
$$\hat{\mu}_{\mathbf{g},\mathbf{h}}(\mathbf{t}) = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.00002 & 0.906444 & 0.093554 \\ 0.00002 & 0.003041 & 0.996957 \end{bmatrix} \end{matrix}$$

We thus get $\mu_{21}(t) = 365 * 0.003041 \approx 1.1$ and $\mu_{12}(t) = 365 * 0.093554 \approx 34$. The time scale we are using is $t = \text{days}$, which means that $\mu_{21}(t)$ tells us that our clients get sick once per year and $\mu_{12}(t)$ tells us that once they are sick, they are sick for around 11 days, which is equal to $1/34$ of a year. Our next step is to evaluate if it is reasonable to assume that the rates are approximately the same over different calendar years. Instead of estimating $\hat{\mu}_{g,h}(t)$ over the entire duration t we calculate $\hat{\mu}_{g,h}(t)$ daily, monthly and yearly. We start with $\hat{\mu}_{1,2}(t)$.

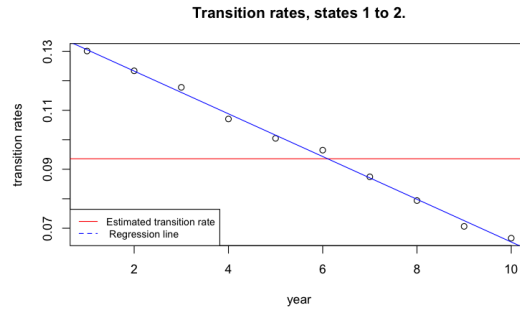
We can clearly see from figure 5 that $\mu_{1,2}(t)$ linearly decreases as an individual gets older, this makes perfect sense. As our customers age, they spend more days being sick. By



(a) intensities aggregated by day



(b) Intensities aggregated by month

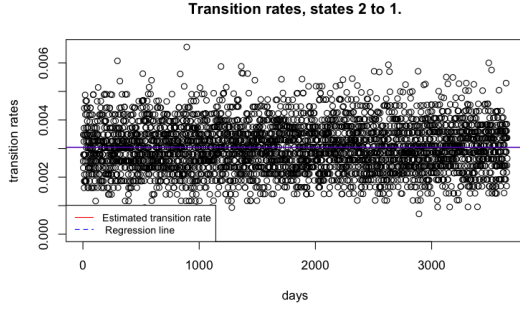


(c) Intensities aggregated by year

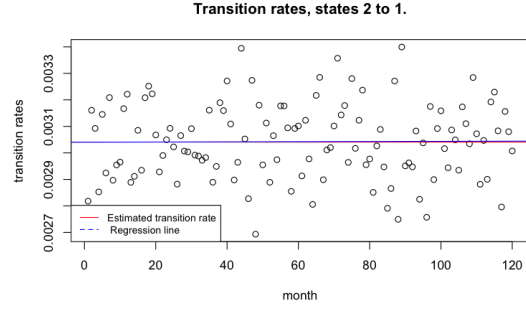
Figure 5: $\mu_{1,2}(t)$ day, month and year.

assuming that $\mu_{1,2}(t)$ linearly decreases, we can, for instance, approximate future $\mu_{1,2}(t)$ with the help of a linear function, $\mu_{1,2}(t) = a - c \cdot t$ where a and c are constants that t is time.

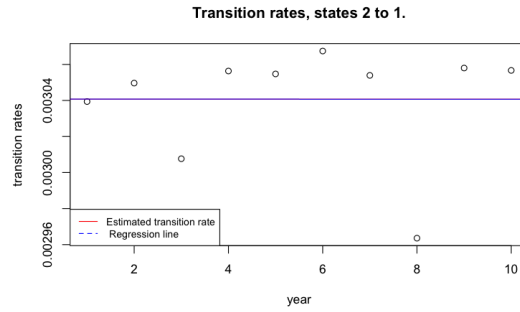
For $\hat{\mu}_{2,1}(t)$ we get the following figures.



(a) intensities aggregated by day



(b) Intensities aggregated by month



(c) Intensities aggregated by year

Figure 6: $\mu_{2,1}(t)$ day, month and year.

There are two big dips at year 3 and 8, we are not sure why. This means that we see a decrease in customers getting sick. Otherwise we draw the conclusion that being healthy to sick stays constant over the years.

4 Assignment 3

For assignment 3, we need to determine the single fair premium to be paid from the time point when a healthy individual turns 50. We are supposed to do these calculations using a discretization corresponding to the time units 1 day, 1 month, and 1 year. To calculate the single fair premium, we need to solve the differential equation,

$$\frac{d}{dt}V^j(t) = r(t)V^j(t) - b^j(t) - \sum_{k \neq j} \mu_{jk}(t)(b^{jk}(t) + V^k(t) - V^j(t)) \quad (6)$$

With the following assumptions, $b^{jk}(t) = 0$ the annual sickness benefit is only paid in state sick and not in transitions. We also assume that Jesus is not in the data and as such $V^0(t) = 0$. This leads us to solve the following system of differential equations,

$$\begin{aligned} \frac{d}{dt}V^1(t) &= r(t)V^1(t) - b^1(t) + \mu_{10}(t)V^1(t) - \mu_{12}(t)(V^2(t) - V^1(t)) \\ \frac{d}{dt}V^2(t) &= r(t)V^2(t) - b^2(t) + \mu_{20}(t)V^2(t) - \mu_{21}(t)(V^1(t) - V^2(t)). \end{aligned} \quad (7)$$

The term $b^2(t)$ is unknown and will be set to 0 and note that the single premium is given by $V^j(0-) = 0$. We say that the contract is actuarial fair as well. This means that prior to initiation of the contract, the benefits (positive increments of $b^j(t)$) and premiums (negative increments of $b^j(t)$) must fulfill this single equation. The results are given in Table 5 and graphically in Figure 7.

Tabell 5: Reserve and single premium

Time discritisation	V^1	V^2
Day	29509.43	28383.86
Month	29509.7	28384.13
Year	26836.60	25310.19

Being able to adjust a portfolio is important as things change. Choosing a smaller time step allows for more adjustment points and, in theory, a better estimation of the true value. From Table 5, we would conclude that we underestimate the true value by using a time discretization in years.

Looks can be deceiving! From Figure 7, it is difficult to tell the difference between each time discretization, yet there is a difference, as shown in Table 5. By increasing the time step, we smooth out the values, allowing for (in theory) a better estimation of the true value.

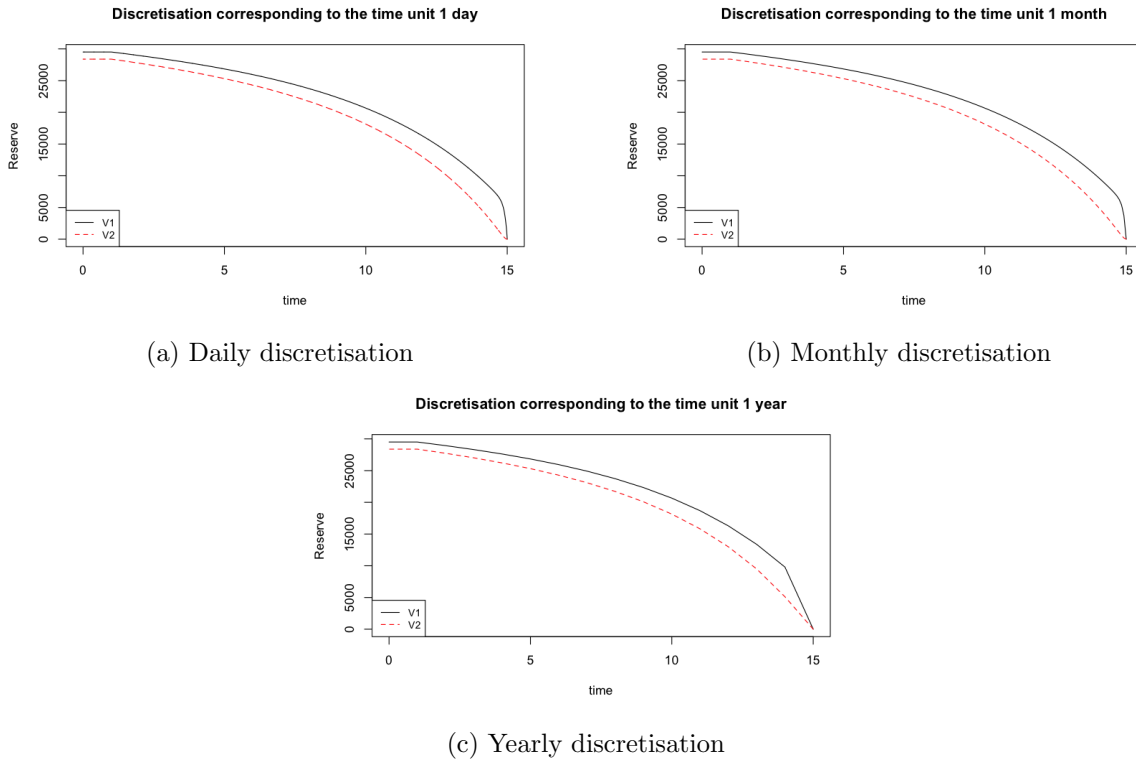


Figure 7: Different time discretisation

5 Assignment 4

In assignment 4 we will redo (A.3) when using a premium paid continuously in time as long as an individual is healthy, which is increased with 1.5% per annum, we choose a daily time discretisation. We will only allow a non-zero premium for the first 10 years to illustrate the effect of the premium payment structure. With the help of looping over a vector filled with different values of b^2 to solve $V^1(0) = V^2(0) = 0$ we get $b^2(0) = -2344.148$. The result of the development is captured in Figure 8. Again, we have a situation where looks can be deceiving or just subpar programming skills in Rstudio. Figure 8 shows a odd mismatch between time zero to one which is not reflected by the vector over the development, see attached code.

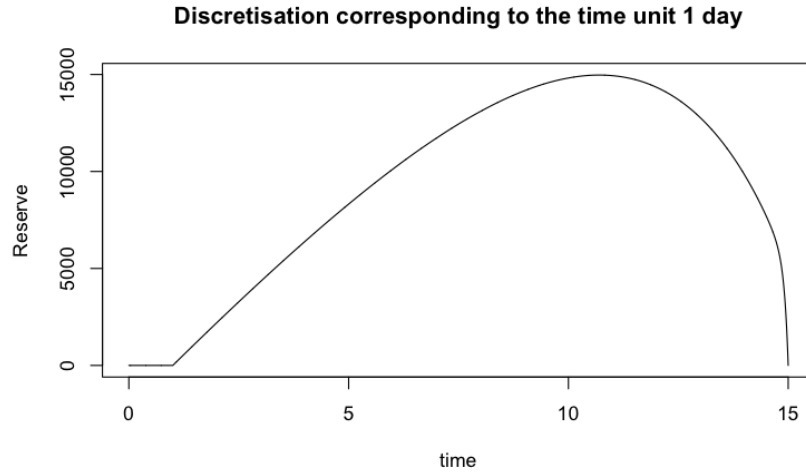


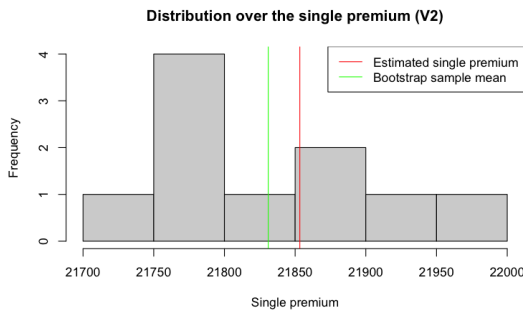
Figure 8: Premium payment structure development

6 Assignment 5

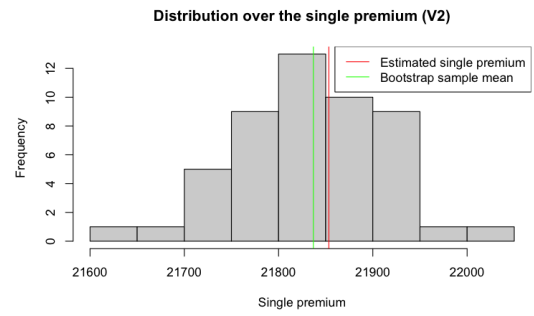
From the 1.1 summary, we concluded that our data underestimates death compared to the true population, and a solution would be to bootstrap our data. Bootstrapping is done by repeatedly sampling (with replacement) the sample data set to create many simulated samples. Each simulated bootstrap sample is used to calculate the fair single premium, and these estimates can then be used to display a sampling distribution. How many samples are enough? There is no definitive answer, but it is assumed that more samples are better. By increasing the number of bootstraps, we reduce the potential effects of random sampling errors possibly caused by the bootstrapping itself. We will, therefore, use and compare 10, 50, and 500 samples. Bootstrapping will not magically add information to the data, but if our sample is from the true population, like the data from SCB, we would assume that by bootstrapping our sample data, we will improve our estimates to reflect the true population better.

We have a very large data set combined with many functions for calculating the estimates. To bootstrap 50 to 500 samples, we need to simplify the functions used. By just assuming that both $\hat{\mu}_{2,1}(t)$ and $\hat{\mu}_{1,2}(t)$ are constant, we drastically reduce the system time needed for calculating the larger samples.

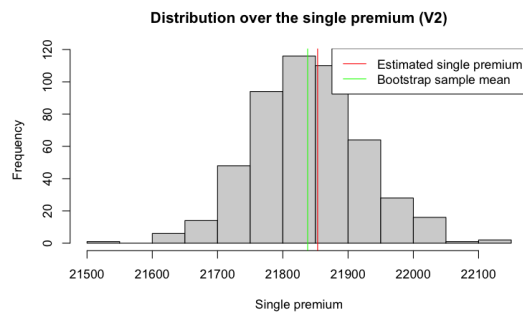
From Figure 8, we can clearly see that at every sample size, we consistently overestimate the fair single premium. What are the implications of this, from assignment 3 we defined mathematically the condition for the single fair premium and calculated it as such yet our bootstrap has shown us that this is overestimated, which would mean that in practice our single fair premium is not fair?



(a) 10 Bootstraps



(b) 50 Bootstraps



(c) 500 Bootstraps

Figure 9: Bootstrap: different sample sizes

7 Plots, more plots!

8 References

<https://www.statistikdatabasen.scb.se>

<https://www.scb.se/hitta-statistik/sverige-i-siffror/manniskorna-i-sverige/doda-i-sverige/#doda-efter-alder>

Norberg R. (2002) Basic Life Insurance Mathematics, lecture notes.